

Melvin Osei Opoku

Gainesville, FL 32612 | melvinoseiopoku@ufl.edu | +1 (678) 303-6791 |
<https://www.linkedin.com/in/melvin-osei-opoku/> | melvinoseiopoku.com

EDUCATION

Bachelor of Science in Biomedical Engineering, Minor in

May 2026

Electrical Engineering

GPA: 3.83/4.0

University of Florida Honors Program, Gainesville, FL

WORK EXPERIENCE

Undergraduate Research Assistant

August 2023 – Present

Brain Mapping Lab, University of Florida, Gainesville

- Developed multimodal pipelines (EMG, audio, video), improving tic detection by 70%
- Trained a machine learning algorithm for automated tic detection in Tourette patients
- Co-authored a peer-reviewed paper on a multimodal tic detection pipeline [1]
- Presented research at the BMES 2024 Annual Meeting

Product Designer

August 2025 – Present

Exactech Inc | Gainesville, FL

- Designed a custom PCB-based bioimpedance measurement system using AD5941 and STM32G0
- Completed schematic design, PCB layout, soldering, firmware bring-up, and hardware validation
- Wrote Embedded C firmware for SPI/I2C, OLED, GPIO, reset, and interrupt testing
- Debugged power, communication, and mixed-signal behavior during board bring-up

Research Intern

June 2025 – August 25

Lewis Lab, Massachusetts Institute of Technology, Cambridge

- Segmented HERCULES-edited MRS data using EEG-based sleep stage classification
- Worked on quantifying changes in 13 brain metabolites across healthy and depressed subjects
- Assisted in overnight polysomnography and MRI data acquisition in human studies

Research Intern

June 2024 – August 24

Murthy Lab, Princeton University, Princeton

- Analyzed FlyWire connectome data on Drosophila LC11 neurons for structure-function relationships
- Discovered inhibitory synapse predominance, challenging the center-surround antagonism model
- Shadowed split-GAL4 genetic experiments to complement computational work
- Presented findings at Society for Neuroscience (SfN) 2024

Research Intern

January 2023 – December 23

American Society of Pharmacognosy, University of Florida, Gainesville

- Expressed and purified 32 UbiA terpene synthases in a GGPP-E. coli overproduction system
- Applied molecular docking and mutagenesis to characterize activity
- Co-authored 3 publications in Nature[2], Elsevier[3], and ACS Catalysis[4]
- Presented findings at the 2023 American Society for Pharmacognosy Annual Meeting

Resident Assistant

August 2023 – May 2025

Housing and Residential Life, University of Florida | Gainesville, FL

- Devised events to encourage community belonging among 32 residents
- Enforced community standards and policies to ensure a safe and supportive living environment
- Served in an on-call rotation by responding to emergencies and crises

LEADERSHIP & SERVICE

Product Designer

August 2023 – May 2024

Generational Relief in Prosthetics, University of Florida | Gainesville, FL

- Designed and optimized Unlimbited prosthetic arm using SolidWorks
- 3D printed and conducted non-destructive testing on a prosthetic arm

Director of Interfaith Affairs

August 2022 – May 2024

Student Government, University of Florida | Gainesville, FL

- Program interfaith events to encourage a campus environment open to students of all faiths
- Ensure a cohesive relationship between the 9 religious groups on campus

Undergraduate Teaching Assistant

August 2023 – December 2023

Secrets of Alchemy, University of Florida | Gainesville, FL

- Graded weekly lab reports and provided helpful feedback
- Held office hours for a class of 60 students: increasing grade by 20%
- Assisted the instructor in preparing the lab, supporting students during the lab, and cleaning up

SKILLS

- **Programming:** NEURON, Python, MATLAB, C++, R, HTML
- **Hardware:** Embedded C, STM32, AD5941, SPI/I2C, PCB Design (KiCad), Prototyping, Circuit Debugging
- **Design & Analysis:** MRS, SolidWorks, Fusion 360, Connectomics, AutoDock Vina, Biostatistics, EEG sleep scoring
- **Laboratory:** MRI, EEG, DMA, TMA, PCR, Genetic Mutation, NMR data analysis, Molecular Docking, GC-MS data analysis
- **Software:** Microsoft Office, Adobe, Label Studio, YOLOv8
- **Language:** Fluent: English, Twi | Conversational: French

AWARDS/SCHOLARSHIP

- | | |
|--|------------------------|
| • Undergraduate Student Excellence Award- UF BME | December 2025 |
| • AI Scholars Program- \$1750 research stipend | August 2025 – May 2026 |
| • University Scholars Program- \$1750 research stipend | August 2024 – May 2025 |
| • Certificate of Outstanding Merit from the College of Engineering | November 2023 |
| • UF Hamilton Center Society Fellow- \$2,500 stipend and trip to Oxford and Cambridge University | August 2023 |
| • American Society of Pharmacognosy - \$5000 research stipend | May 2023 –August 2023 |
| • Wentworth Travel Scholarship – \$500 Research Travel Funding | May 2023 – August 2023 |
| • Emerging Scholar Award- \$1000 research stipend | January 2023 |
| • Davis United World Scholar- \$160,000 fully funded University of Florida undergraduate degree | August 2022 – May 2026 |

PUBLICATIONS

- [1] G. Lowor, M. Osei Opoku, *et al.*, “A comparative study of video-based and electromyography-based detection of tics,” *Clin. Park. Relat. Disord.*, vol. 14, p. 100438, 2026, doi: 10.1016/j.prdoa.2026.100438.
- [2] X. Wei *et al.*, “Exploring and expanding the natural chemical space of bacterial diterpenes,” *Nat. Commun.*, vol. 16, no. 1, p. 3721, Apr. 2025, doi: 10.1038/s41467-025-57145-6.
- [3] T. A. Alsup, M. Osei Opoku, and J. D. Rudolf, “Characterization of UbiA terpene synthases with a precursor overproduction system in *Escherichia coli*,” in *Methods in Enzymology*, vol. 699, Elsevier, 2024, pp. 395–417. doi: 10.1016/bs.mie.2024.02.001.
- [4] T. A. Alsup *et al.*, “Discovery of UbiA-Type Cyathane Synthases in Bacteria,” *ACS Catal.*, vol. 15, no. 19, pp. 16873–16881, Oct. 2025, doi: 10.1021/acscatal.5c04650.